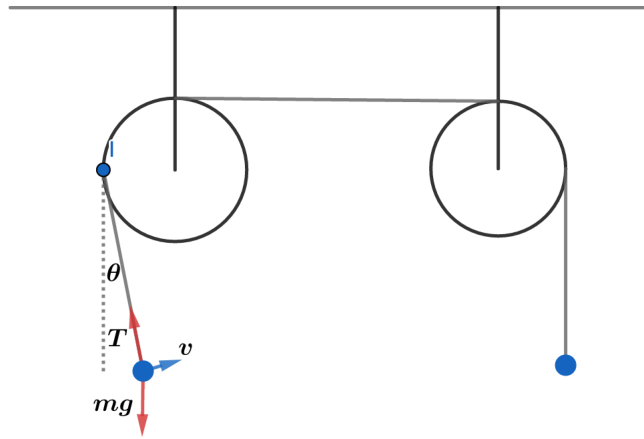


## 2018B F=ma Exam: Problem 22

Kevin S. Huang



Suppose the mass on the right remains at rest. Then the left mass would swing like a pendulum by conservation of string. For circular motion,

$$T = mg \cos \theta + \frac{mv^2}{R}$$

so the tension oscillates. Averaging over time, the vertical component of tension supports the weight of the mass while the horizontal component of tension provides the restoring force. Thus, the average tension is greater than  $mg$ .

Hence, the right mass must oscillate vertically and move upward so the answer is B.